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On the Fast Track: Information Acquisition Costs and Information Production

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1 Big picture

- **Question.** Does lowering information-acquisition costs cause agents to collect more information, produce higher-quality information, and improve price efficiency? The paper studies this question using the expansion of China's high-speed rail network as a shock to travel costs faced by sell-side analysts.
- **Economic setting.** The agents are Chinese sell-side analysts. The objects they study are listed firms. The key input into analyst research is firm-specific information, especially soft information obtained through site visits, face-to-face meetings, facility tours, and interactions with managers and non-management employees.
- **Main friction.** Before a high-speed rail connection, visiting a distant portfolio firm is costly in time, uncertainty, and scheduling flexibility. If acquisition costs matter, analysts should visit more once rail connections make travel easier.
- **Core shock.** A firm-brokerage pair becomes treated when at least one high-speed rail route connects the firm's city to the brokerage city's location. The paper lags the connection by one year, so the effect is measured after the connection has had time to become useful for analyst travel.
- **Identification idea.** The design compares the same firm-brokerage pair before and after it becomes connected by HSR, relative to firm-brokerage pairs that are not yet connected. The preferred specifications include firm-brokerage-pair fixed effects and city-by-year fixed effects; a stronger specification includes firm-year and broker-year fixed effects.
- **Main result on information collection.** HSR introduction increases the frequency of analyst site visits. In the baseline site-visit regression, the HSR coefficient is 0.049, which means treated analyst teams increase site visits by about 4.9% relative to controls.
- **Main result on information quality.** HSR introduction improves analyst forecast quality. In the baseline forecast-error regression, the HSR coefficient is -0.021, which implies analyst forecast errors decline by about 2.1% after the HSR connection.
- **Main result on price formation.** Lower acquisition costs make analyst outputs more informative to the market. Post-HSR, market reactions to forecast revisions rise by about 1.7%, and market reactions to stock recommendations rise by about 1.9%. At the same time, market reactions to firms' earnings announcements fall when a larger share of analysts covering the firm are HSR-connected.
- **Mechanism.** The evidence points to soft information. The HSR effect is stronger when the firm is difficult to visit without HSR, when soft information is more important for valuation, when the firm is informationally opaque, and when governance is weaker.
- **Survey evidence.** The paper surveys 334 financial analysts. Among remote analysts, about 96% agree that a direct HSR connection leads analysts to conduct more site visits, about 82% say it helps them make more accurate earnings forecasts, and roughly 87% agree that it helps them understand soft-information items such as culture, strategy, operations, challenges, and employee morale.

- **Takeaway.** The paper provides direct micro evidence for a foundational idea in information economics: when information becomes cheaper to acquire, informed agents collect more of it, produce better output, and help prices incorporate information faster.

2 Data and measurement

2.1 Institutional background: China's HSR network

China's Ministry of Railway announced a high-speed rail plan in 2006. The State Council later set the objective of building a national HSR grid linking provincial capitals and major cities. The network expansion created a staggered source of variation in travel costs across city pairs.

The paper emphasizes three reasons why HSR reduces analysts' information-acquisition costs:

- **Lower travel time.** For example, Beijing-Shanghai travel time fell from roughly 13 hours by traditional train to about five hours by HSR.
- **Lower travel uncertainty.** HSR is less exposed than flights to weather and airport delays and is often better connected to inner-city transit.
- **Greater scheduling flexibility.** HSR routes often run frequently, which makes it easier for analysts to schedule site visits around firm events and other coverage responsibilities.

The HSR network therefore changes the cost of collecting face-to-face information, not merely the speed at which public information travels.

2.2 Main data sources

The paper combines several data sources:

- **CSMAR analyst data:** analyst forecasts, stock recommendations, stock prices, and firm financial information for Chinese listed firms.
- **CNRDS site-visit data:** corporate site visits disclosed by Shenzhen Stock Exchange firms. These data are available because SZSE began requiring disclosure of corporate site visits, first in annual reports and later through real-time disclosure.
- **China Railway Yearbooks:** manually collected information on HSR routes, stops, and opening dates from 2003 to 2019.
- **Survey data:** a large-scale survey of financial analysts about HSR, site visits, information collection, and forecast quality.

2.3 Sample construction

The main analyst-forecast sample covers 2005–2019. The authors exclude financial firms, firms on the Growth Enterprise Market, special-treatment firms, IPO-year observations, and broker-firm pairs covered for less than one year. The main sample also excludes firms in mega cities and local analysts because the treatment is meant to capture reductions in travel costs for nonlocal

information acquisition.

The final analyst-forecast sample contains:

- 161,858 annual EPS forecasts;
- 11,381 firm-years;
- 1,596 distinct firms in 243 cities;
- 82 brokerages located in 33 cities.

The site-visit sample covers 2009–2019 because site-visit disclosure is available for Shenzhen Stock Exchange firms. It contains:

- 33,200 firm-broker-year observations;
- 858 distinct firms located in 191 cities;
- 80 brokerages located in 33 cities.

2.4 Treatment variable

The key treatment variable is

$$\text{HSR}_{a,i,t} = 1$$

if at least one HSR route connects brokerage city a and firm city i at the beginning of year t , and zero otherwise.

The paper constructs the indicator so that if the route is introduced during year $t - 1$, $\text{HSR}_{a,i,t} = 1$ in year t and after. This lag is useful because analysts need time to adjust travel schedules and information-production routines.

2.5 Information-collection outcome

The main direct measure of information collection is the number of site visits conducted by a brokerage for a firm in a year. The regression outcome is

$$\text{Visits}_{a,i,t} = \log(1 + \# \text{Analyst Site Visits}_{a,i,t}).$$

A site visit usually occurs at the firm's headquarters and can involve meetings with managers, investor-relations officers, employees, and tours of production or operating facilities.

2.6 Forecast-quality outcome

Analyst information quality is measured using forecast errors. For forecast f issued by analyst team or brokerage a for firm i , the basic forecast-error measure is

$$\text{FE}_{a,i,f,t} = \log \left(1 + 100 \cdot \frac{|\widehat{\text{EPS}}_{a,i,f,t} - \text{EPS}_{i,t}|}{P_{i,t-1}} \right),$$

where $\widehat{\text{EPS}}$ is the analyst EPS forecast, EPS is actual EPS, and $P_{i,t-1}$ is the share price at the beginning of the year.

The use of absolute error makes smaller values better. The log transformation reduces the effect of skewness.

2.7 Price-efficiency outcomes

The paper studies three price-formation outcomes:

- **Forecast-revision information, FRINFO.** This is the market reaction to analyst forecast revisions, measured using three-day size-adjusted cumulative abnormal returns around the revision.
- **Recommendation information.** This is the market reaction to analyst stock recommendations, again measured with short-window abnormal returns.
- **Earnings-announcement information, EAINFO.** This is the absolute three-day size-adjusted abnormal return around the annual earnings announcement.

The economic logic is simple. If analysts produce and disseminate more information before earnings announcements, the market should react more to analysts' releases and less to the firm's later earnings announcement.

2.8 Difficult-to-visit indicator

The paper uses a stress-test variable:

$$\text{Difficult to visit}_{a,i} = 1$$

if the distance between the firm and brokerage exceeds 800 km and no direct flight existed between the cities by 2005. This variable is measured before the main HSR expansion. If the HSR effect is truly about lower travel costs, it should be stronger for these difficult-to-visit pairs.

2.9 Soft-information measure

The paper constructs a firm-level measure of whether soft information is important for forecasting the firm. The idea is that distance should matter more when soft information matters. The authors compare forecast accuracy of nearby analysts and distant analysts over the previous three years:

$$\text{SoftInfo}_{i,t} = 1 \quad \text{if} \quad \overline{\text{FE}}_{\text{near},i,t-3:t-1} < \overline{\text{FE}}_{\text{distant},i,t-3:t-1}.$$

If near analysts are more accurate than distant analysts, the firm is classified as one for which soft information is important.

3 Models and empirical specifications

3.1 Main difference-in-differences model

The main specification is

$$y_{a,i,t} = \beta_0 + \beta_1 \text{HSR}_{a,i,t} + \mathbf{X}'_{a,i,t} \Gamma + \alpha_{a,i} + \lambda_{\text{city}(i),t} + \lambda_{\text{city}(a),t} + \varepsilon_{a,i,t}. \quad (1)$$

Here:

- a indexes the brokerage or analyst team;
- i indexes the firm;
- t indexes the year;
- $y_{a,i,t}$ is the outcome, such as site visits, forecast errors, or market reactions to analyst outputs;
- $\alpha_{a,i}$ is a firm-brokerage-pair fixed effect;
- $\lambda_{\text{city}(i),t}$ is a firm-city-by-year fixed effect;
- $\lambda_{\text{city}(a),t}$ is a brokerage-city-by-year fixed effect;
- $\mathbf{X}_{a,i,t}$ contains controls such as firm size, leverage, ROA, book-to-market, return volatility, analyst following, and analyst firm-specific experience.

The coefficient of interest is β_1 . For site visits, a positive β_1 means HSR induces more information collection. For forecast errors, a negative β_1 means HSR improves the quality of analyst information production.

3.2 Stronger fixed-effect specification

A stronger version replaces city-year fixed effects with firm-year and brokerage-year fixed effects:

$$y_{a,i,t} = \beta_0 + \beta_1 \text{HSR}_{a,i,t} + \mathbf{X}'_{a,i,t} \Gamma + \alpha_{a,i} + \delta_{i,t} + \mu_{a,t} + \varepsilon_{a,i,t}. \quad (2)$$

This is powerful because it absorbs time-varying shocks specific to a firm and time-varying shocks specific to a broker. Identification comes from variation across brokers covering the same firm-year and across firms covered by the same broker-year.

3.3 Heterogeneity by difficulty of visiting

To test whether HSR matters more when travel constraints were binding, the paper estimates

$$y_{a,i,t} = \beta_0 + \beta_1 \text{HSR}_{a,i,t} + \beta_2 (\text{HSR}_{a,i,t} \times \text{Difficult}_{a,i}) + \mathbf{X}'_{a,i,t} \Gamma + \alpha_{a,i} + \lambda_{\text{city}(i),t} + \lambda_{\text{city}(a),t} + \varepsilon_{a,i,t}. \quad (3)$$

The key test is β_2 . A positive β_2 for visits or a negative β_2 for forecast errors supports the interpretation that reduced travel costs drive the results.

3.4 Price reaction to analyst outputs

For forecast revisions and recommendations, the paper uses the main DiD model in Eq. (1), with the dependent variable set to FRINFO or the market reaction to stock recommendations. The prediction is

$$\beta_1 > 0.$$

That is, once analysts have easier access to firms, the market should react more strongly to their research because the research contains more value-relevant information.

3.5 Price reaction to earnings announcements

For earnings announcements, the unit of analysis is the firm-year. The paper constructs PHSR, the intensity of analyst HSR connections to a firm, and estimates

$$\text{EAINFO}_{i,t} = \theta_0 + \theta_1 \text{PHSR}_{i,t} + \mathbf{Z}'_{i,t} \Pi + \alpha_i + \delta_t + u_{i,t}. \quad (4)$$

The paper uses two versions of PHSR:

- PHSR1: the proportion of analysts covering the firm who are HSR-connected to the firm;
- PHSR2: the proportion of research reports signed by HSR-connected analysts.

The predicted sign is $\theta_1 < 0$. If analysts reveal more information before the earnings announcement, then the earnings announcement contains less new information.

3.6 Two-stage least squares: site visits as the channel

The reduced-form DiD estimates show that HSR affects both visits and outcomes. The paper also asks how much the outcomes change when site visits themselves increase. It instruments site visits using HSR.

First stage:

$$\text{Visits}_{a,i,t} = \pi_0 + \pi_1 \text{HSR}_{a,i,t} + \mathbf{X}'_{a,i,t} \Phi + \alpha_{a,i} + \lambda_{\text{city}(i),t} + \lambda_{\text{city}(a),t} + v_{a,i,t}. \quad (5)$$

Second stage:

$$\text{Output}_{a,i,t} = \kappa_0 + \kappa_1 \widehat{\text{Visits}}_{a,i,t} + \mathbf{X}'_{a,i,t} \Psi + \alpha_{a,i} + \lambda_{\text{city}(i),t} + \lambda_{\text{city}(a),t} + e_{a,i,t}. \quad (6)$$

The output variables are forecast error, market reaction to forecast revisions, and market reaction to stock recommendations.

3.7 Coverage-initiation model

To study the extensive margin of information production, the paper estimates

$$\text{Initiation}_{a,i,t} = \rho_0 + \rho_1 \text{HSR}_{a,i,t} + \mathbf{X}'_{i,t} \Omega + \alpha_{a,i} + \lambda_{\text{city}(i),t} + \lambda_{\text{city}(a),t} + \eta_{a,i,t}. \quad (7)$$

Initiation $_{a,i,t} = 1$ if the broker issues a forecast for firm i in year t after issuing no forecast for that firm during years $t - 3$ through $t - 1$. A positive ρ_1 means HSR increases the likelihood that analysts start covering a new firm.

4 Identification and instruments

4.1 What identifies the HSR effect

The central identifying variation is within firm-brokerage-pair changes in HSR connectivity over time. The pair fixed effect removes stable distance, stable information asymmetry, stable brokerage skill, and persistent firm-broker matching. City-year fixed effects remove shocks common to all firms in a city-year or all brokerages in a city-year.

The main comparison is therefore not simply connected cities versus unconnected cities. It is whether a given firm-broker pair changes outcomes after the pair becomes connected, relative to other pairs facing the same city-level time shocks.

4.2 Why the fixed effects matter

HSR construction is not randomly assigned. Fast-growing cities may be more likely to receive HSR lines. A broker located in a growing financial center may also change analyst behavior for reasons unrelated to HSR. The paper addresses this through several layers of fixed effects:

- **Pair fixed effects** control for persistent pair-level relations.
- **Firm-city-by-year fixed effects** control for shocks to the firm city's economy or information environment.
- **Brokerage-city-by-year fixed effects** control for shocks to the broker city's analyst labor market or brokerage industry.
- **Firm-year and broker-year fixed effects** in the stronger specification absorb time-varying shocks at the individual firm and individual broker level.

The firm-year and broker-year specification is especially demanding because it compares treated and untreated relationships within the same firm-year and within the same broker-year.

4.3 Endogenous HSR construction and lobbying concerns

A major concern is that cities or firms may lobby for HSR connections because of expected economic growth or rising investor demand. If this lobbying is correlated with analyst activity, HSR would not be a clean shock.

The paper addresses this concern in two ways:

- It excludes firm-broker pairs located at the two ends of an HSR line. This focuses on pairs connected as intermediate city pairs rather than likely targets of endpoint lobbying.
- It excludes the largest 5% of firms by assets because large firms are more likely to have political

influence or resources to lobby for infrastructure.

The main results remain robust in these exercises.

4.4 Placebo tests

The main sample excludes mega-city firms and local analysts because HSR should matter less when travel was already easy. The paper then uses these excluded observations as a placebo sample. HSR has no significant effect on site visits or forecast errors in this group.

This is important because it shows the estimated effect is not merely a broad time trend in analyst work or a generic improvement in China's capital markets.

4.5 Parallel trends

The paper estimates event-study versions of the main model. The event-study coefficients show no meaningful pretrend in site visits or forecast errors before HSR introduction. The effect appears after HSR becomes operational and is stronger one or more years after the connection.

This supports the DiD interpretation: treated and control pairs were not already diverging before the HSR connection.

4.6 Soft-information mechanism as a credibility check

The soft-information tests strengthen identification because they examine where the effect should be strongest. If HSR simply captures city growth, one would not necessarily expect the effect to be concentrated among hard-to-visit firms, soft-information-intensive firms, high-volatility firms, or weak-governance firms.

The fact that the effect is concentrated in these groups supports the paper's mechanism: HSR changes the cost of collecting hard-to-transfer, face-to-face information.

4.7 Role of the survey evidence

The survey is not the primary identification strategy, but it complements the regressions. Analysts report that direct HSR connections make them visit more often, make visits more flexible, and help them understand firm-specific soft information. The remote-analyst responses are stronger than the mega-city-analyst responses, which matches the empirical heterogeneity.

This survey evidence reduces the concern that the empirical results are driven by an unobserved mechanical association unrelated to analyst information acquisition.

4.8 Remaining limitations

The paper cannot literally randomize HSR lines. The empirical design therefore relies on fixed effects, event studies, placebo samples, and robustness tests. Also, site-visit data are available mainly for SZSE firms, so some analyses necessarily use a narrower sample. Finally, the paper studies China and sell-side analysts; the mechanism should generalize to other settings where soft information and travel costs matter, but the magnitudes may differ.

5 Tables and figures

5.1 Figure 1: National development of China's HSR network

Read. Figure 1 documents the rapid expansion of China's HSR network. The paper describes four stages: no operation before August 2008, preliminary operation in 2008–2011, steady network expansion in 2011–2013, and rapid expansion through 2018.

Economic message. The institutional setting provides staggered variation in travel costs across city pairs. This is exactly the type of variation needed to test whether lower information-acquisition costs change analyst behavior.

The table and figure depict the introduction and development of high-speed rail (HSR) in China. The figure was reproduced following Figure 2 of Huang et al. (2018) with an extension of HSRs completed between August 2017 and December 2018.

Time Period	HSR Lines	Operating mileage (km)
Before August 2008	Conventional railway network, no HSR service	0
Between August 2008 and July 2011	Preliminary construction	6963
Between July 2011 and January 2013	Network skeleton: Several new HSR lines were built	12344
Between January 2013 and December 2018	Deep intensification	290000



Fig. 1. National development of HSR network in China.

Table 1
Summary statistics.
This table provides summary statistics for variables used in site-visits analyses in Panel A, forecast-errors analyses in Panel B, and stock-price-efficiency analyses in Panel C. HSR is an indicator variable that takes the value of 1 for year t if at least one HSR connects the firm city and the brokerage city at the beginning of t , and 0 otherwise. Other variable definitions are provided in Appendix A, Table A1

Panel A				
Variable	# of Obs.	Mean	S.D.	Median
# of Analyst Site Visits	33,200	0.590	0.964	0.000
HSR	33,200	0.405	0.491	0.000
Total assets (Billion Yuan)	33,200	18.814	36.038	6.606
Leverage	33,200	0.431	0.188	0.423
ROA	33,200	0.070	0.062	0.000
Book-to-market	33,200	0.419	0.308	0.341
Return volatility	33,200	0.119	0.066	0.109
# of analysts following	33,200	15.307	8.516	15.000
Firm experience	33,200	2.066	1.496	1.500
Difficult to visit	33,200	0.472	0.499	0.000
Soft information	28,309	0.520	0.500	1.000
Panel B				
Variable	# of Obs.	Mean	S.D.	Median
Analyst-Forecast Error (%)	161,785	1.216	1.666	0.000
Initiation	656,526	0.042	0.201	0.000
HSR	161,785	0.390	0.488	0.000
Total assets (Billion Yuan)	161,785	23.414	41.643	8.731
Leverage	161,785	0.455	0.186	0.457
ROA	161,785	0.074	0.058	0.065
Book-to-market	161,785	0.434	0.322	0.348
Return volatility	161,785	0.119	0.056	0.109
# of analysts following	161,785	16.421	8.766	16.000
Firm experience	161,785	2.147	1.547	1.500
Difficult to visit	161,785	0.469	0.499	0.000
Soft information	137,558	0.527	0.499	1.000
Panel C				
Variable	# of Obs.	Mean	S.D.	Median
Market reaction to analyst-forecast revisions (%) (FRINFO)	53,201	0.916	5.301	0.444
Market reaction to stock recommendations (%) (EAINFO)	52,374	1.212	5.233	0.407
Market reaction to earnings announcements (%) (EAINFO)	21,590	3.805	4.590	2.752
Proportion of HSR-connected analysts (PHSR1)	21,590	0.191	0.343	0.000
Proportion of earnings forecasts issued by HSR-connected analysts (PHSR2)	21,590	0.192	0.349	0.000

5.2 Table 1: Summary statistics

Read. Table 1 reports summary statistics for the site-visit sample, forecast-error sample, and price-efficiency sample. The sample contains nonlocal analyst-firm relationships across many firm and brokerage cities, which is necessary for identifying HSR effects.

Economic message. The table shows the paper has direct measures of the information-production process, not just final market outcomes. This is what makes the paper different from studies that infer information acquisition indirectly.

5.3 Table 2: HSR introduction and analyst site visits

Outcome: log site visits	Baseline	Firm-year/broker-year FE	Difficult-to-visit interaction
HSR	0.049***	0.028**	0.031*
HSR $\times$ Difficult to visit	—	—	0.046*
Observations	31,417	29,977	31,417

Read. The baseline coefficient 0.049 implies that HSR-connected analysts increase site visits by

about 4.9%. The coefficient remains positive and significant with firm-year and brokerage-year fixed effects. The interaction with difficult-to-visit pairs is positive.

Economic message. Lower acquisition costs increase direct information collection. The stronger effect for difficult-to-visit pairs is a key mechanism test: HSR matters most when travel costs were high before the shock.

Table 2

HSR introduction and analyst site visits.

This table examines the effect of HSR introduction between a firm-brokerage pair on analyst site visits. The unit of analysis is at the broker-firm-pair-by-year level. Column (1) includes firm-broker-pair fixed effects and city-times-year fixed effects. Column (2) includes firm-broker-pair fixed effects and firm-year and broker-year fixed effects. Column (3) explores cross-sectional variation based on column (1) model specification. *HSR* is an indicator variable that takes the value of 1 if at least one HSR connects the firm city and the brokerage city at the beginning of t , and 0 otherwise. The dependent variable, *# of Analyst Site Visits*, is the natural logarithm of 1 plus the number of site visits conducted by a brokerage for a firm in year t . *Difficult to visit* is an indicator variable that equals 1 if the distance between the listed firm and the brokerage house is more than 800 km and if no direct flight existed between the firm and the analyst by 2005, and 0 otherwise. Standard errors clustered at the firm level are reported in parentheses. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

	# of Analyst Site Visits		
	(1)	(2)	(3)
HSR	0.049*** (0.014)	0.028** (0.014)	0.031* (0.016)
HSR*Difficult to visit			0.046* (0.023)
Control variables	Yes	Yes	Yes
Broker-firm pair FE	Yes	Yes	Yes
Firm city $\times$ Year FE	Yes	No	Yes
Firm $\times$ Year FE	No	Yes	No
Brokerage city $\times$ Year FE	Yes	No	Yes
Brokerage $\times$ Year FE	No	Yes	No
Observations	31,417	29,977	31,417
R-squared	0.551	0.673	0.551

Table 3

HSR introduction and analyst-forecast errors.

This table examines the effect of HSR introduction between a firm-brokerage pair on analyst-forecast errors. The unit analysis is at the forecast level. Column (1) includes firm-broker-pair fixed effects and city-times-year fixed effects. Column (2) includes firm-broker-pair fixed effects and firm-year and broker-year fixed effects. Column (3) explores cross-sectional variation based on column (1) model specification. *HSR* is an indicator variable that takes the value of 1 if at least one HSR connects the firm city and the brokerage city at the beginning of t , and 0 otherwise. The dependent variable, *Analyst-Forecast Error*, is the natural logarithm of 1 plus the absolute difference between the EPS forecast and actual EPS, divided by the share price at the beginning of the year and multiplied by 100. *Difficult to visit* is an indicator variable that equals 1 if the distance between the listed firm and the brokerage house is more than 800 km and if no direct flight existed between the firm and the analyst by 2005, and 0 otherwise. Standard errors clustered at the firm level are reported in parentheses. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

	Analyst-Forecast Error		
	(1)	(2)	(3)
HSR	-0.021** (0.009)	-0.011* (0.006)	-0.009 (0.010)
HSR*Difficult to visit			-0.031** (0.014)
Control variables	Yes	Yes	Yes
Broker-firm pair FE	Yes	Yes	Yes
Firm city $\times$ Year FE	Yes	No	Yes
Firm $\times$ Year FE	No	Yes	No
Brokerage city $\times$ Year FE	Yes	No	Yes
Brokerage $\times$ Year FE	No	Yes	No
Observations	160,519	158,842	160,519
R-squared	0.612	0.774	0.612

5.4 Table 3: HSR introduction and analyst forecast errors

Outcome: log forecast error	Baseline	Firm-year/broker-year FE	Difficult-to-visit interaction
HSR	-0.021**	-0.011*	-0.009
HSR $\times$ Difficult to visit	—	—	-0.031**
Observations	160,519	158,842	160,519

Read. Forecast errors fall after HSR introduction. The baseline coefficient implies a 2.1% decline in forecast errors. The effect is much stronger for difficult-to-visit firm-broker pairs.

Economic message. More information collection translates into better information production. The result is not just that analysts travel more; their outputs improve.

5.5 Figure 2: Parallel-trends test

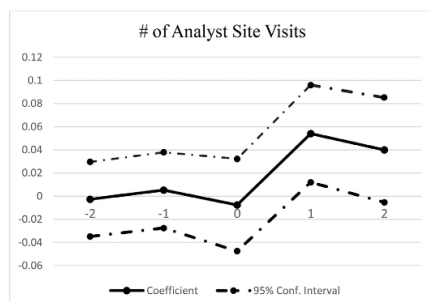
Read. Figure 2 presents event-study coefficients around HSR introduction. The coefficients do not show meaningful differences between treated and control pairs before HSR becomes operational.

The post-treatment effects appear after the rail connection.

Economic message. The figure supports the parallel-trends assumption. It is evidence against the idea that treated pairs were already on a different trajectory before HSR.

Panel A: Testing the parallel-trends assumption for analyst site-visits analysis

Panel A reports the point estimates for *HSR* from an ordinary least square (OLS) regression. The dependent variable is *# of Analyst Site Visits* in logs. The model specification is the same as column (1) of Appendix C. Point estimates are reported for two years before, one year before, the year of, one year after, and two or more years after the introduction of HSR between a firm and its broker. Ninety-five percent confidence intervals, adjusted for clustering at the firm level, are also plotted.



Panel B: Testing the parallel-trends assumption for forecast-errors analysis

Panel B reports the point estimates for *HSR* from an OLS regression. The dependent variable is *Analyst-Forecast Error* in logs. The model specification is the same as in column (2) of Appendix C. Point estimates are reported for two years before, one year before, the year of, one year after, and two or more years after the introduction of HSR between a firm and her broker. Ninety-five percent confidence intervals, adjusted for clustering at the firm level, are also plotted.

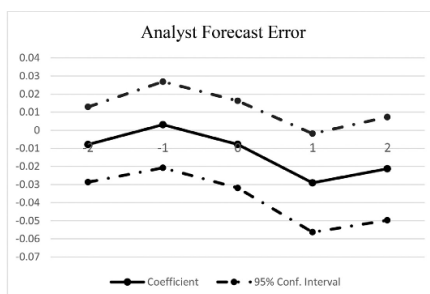


Fig. 2. Testing the parallel-trends assumption.

Table 4

HSR introduction and market reaction to analyst research releases.

This table examines the effect of HSR introduction on the market reaction to analyst research releases, consisting of forecast revisions and stock recommendations. *HSR* is an indicator variable that takes the value of 1 if at least one HSR connects the firm-broker pair at the beginning of *t*, and 0 otherwise. In column (1), the dependent variable, *FRINFO*, is market reaction to analyst-forecast revisions. In column (2), the dependent variable is *Market Reaction to Stock Recommendations*. Both dependent variables are log-transformed with a small constant added to the original variable before log-transformation. Control variables are defined in Appendix A. Table 1. Standard errors clustered at the firm level are reported in parentheses. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

	<i>FRINFO</i> (1)	<i>Market Reaction to Stock Recommendations</i> (2)
HSR	0.017** (0.008)	0.019*** (0.007)
Total assets	0.008 (0.007)	0.007 (0.006)
Leverage	-0.007 (0.026)	0.009 (0.024)
ROA	0.165** (0.069)	0.062 (0.054)
Book-to-market	-0.045*** (0.011)	-0.075*** (0.011)
# of analyst following	-0.010* (0.005)	-0.008 (0.005)
Firm experience	-0.000 (0.005)	-0.004 (0.004)
Forecast duration	0.007* (0.004)	
Broker-firm pair FE	Yes	Yes
Firm city × Year FE	Yes	Yes
Brokerage city × Year FE	Yes	Yes
Observations	47,783	46,948
R-squared	0.297	0.305

5.6 Table 4: HSR introduction and market reaction to analyst research

Outcome	HSR coefficient	Interpretation
Forecast-revision reaction, <i>FRINFO</i>	0.017**	Reaction rises by about 1.7%
Stock-recommendation reaction	0.019***	Reaction rises by about 1.9%

Read. The market reacts more strongly to analyst forecast revisions and recommendations after HSR connection.

Economic message. Investors appear to understand that connected analysts produce more informative research. HSR improves not only private information collection but also public price discovery through analyst releases.

5.7 Table 5: HSR intensity and earnings-announcement information

Outcome: EAINFO	Coefficient	Meaning
PHSR1	-0.039*	More HSR-connected analysts lower EA reaction
PHSR2	-0.045**	More HSR-connected reports lower EA reaction

Read. As the share of HSR-connected analysts or reports increases, the market reaction to earnings announcements falls.

Economic message. Analyst information is incorporated before the earnings announcement. This is consistent with faster price discovery and improved price efficiency.

Table 5

HSR introduction and market reaction to firm earnings announcements.

This table examines the relation between the intensity of analyst HSR connections and the information content of earnings announcements. The unit of analysis is at the firm-year level. Firm fixed effects are included in the estimation. The dependent variable, *EAINFO*, is the natural logarithm of one plus market reaction to earnings announcements, calculated as size-adjusted absolute cumulative abnormal returns for the three-day window $(-1,+1)$ around the annual earnings-announcement date and multiplied by 100. *PHSR1* is the proportion of analysts who are connected to the firm by an HSR at the beginning of t , relative to the total number of analysts covering the firm in year t . *PHSR2* is the proportion of research reports signed by HSR-connected analysts relative to the total number of research reports signed by all analysts covering the firm in t . An analyst is HSR connected if the analyst is connected by an HSR with the portfolio firm at the beginning of t . Control variables are defined in [Appendix A.Table 1](#). Standard errors clustered at the firm level are reported in parentheses. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

	<i>EAINFO</i>	
	(1)	(2)
PHSR1	-0.039* (0.021)	
PHSR2		-0.045** (0.021)
Total assets	-0.019* (0.010)	-0.018* (0.010)
Leverage	-0.002*** (0.000)	-0.002*** (0.000)
ROA	0.000*** (0.000)	0.000*** (0.000)
Book-to-market	-0.053* (0.030)	-0.053* (0.030)
# of analysts following	0.012 (0.009)	0.013 (0.009)
Firm FE	Yes	Yes
Year FE	Yes	Yes
Observations	21,533	21,533
R-squared	0.140	0.140

Table 6

The elasticity of outcomes to site visits—two-stage least-squares estimation.

This table reports the results of two-stage least-squares estimation for the relation of analyst site visits with forecast error, market reaction to forecast revisions, and market reaction to analyst recommendations. The samples contain firms traded at SOE only due to the availability of site-visits data. The dependent variable in the first-stage regression is the natural logarithm of one plus the number of analyst site visits in year t . Column (1) reports the first-stage results when analyst-forecast error serves as the dependent variable in the second stage. Columns (2)–(4) report the results of the second-stage regression, where the dependent variable is Analyst-forecast Error, market reaction to analyst-forecast revisions (*FRNRO*), and market reaction to recommendations, respectively, with all in log-transformation. The explanatory variable is the Predicted # of Analyst Site Visits obtained from the first stage. Standard errors clustered at the firm level are reported in parentheses. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

	First Stage	Second Stage		
	# of Analyst Site Visits (1)	Analyst-Forecast Error (2)	FRNRO (3)	Market Reaction to Stock Recommendations (4)
HSR	0.066*** (0.018)			
Predicted # of Analyst Site Visits		-0.359** (0.174)	0.456* (0.046)	0.409** (0.040)
Control variables	Yes	Yes	Yes	Yes
Broker-firm pair FE	Yes	Yes	Yes	Yes
Firm city × Year FE	Yes	Yes	Yes	Yes
Brokerage city × Year FE	Yes	Yes	Yes	Yes
Observations	79,366	79,366	23,356	22,931
R-squared	0.627	0.718	0.304	0.317

Table 7
The effect of HSR introduction and the importance of soft information.
This table examines the heterogeneous effect of HSR introduction on analyst site visits and forecast errors based on the importance of a firm's soft information for performance valuation. HSR is an indicator variable that takes the value of 1 if at least one HSR connects the firm-broker pair at the beginning of t , and 0 otherwise. The importance of soft information has two proxies. The first is *Soft information*, an indicator variable taking the value of 1 if the average EPS forecast error for the near analyst group is lower than that for the distant analyst group for a given firm over year $t-3$ to year $t-1$. The second proxy is *High return volatility*, an indicator variable taking the value of 1 if the firm's stock return volatility is above the sample median in the year immediately before HSR introduction. In columns (1) and (3), the dependent variable, *# of Analyst Site Visits*, is the natural logarithm of 1 plus the number of site visits conducted by a brokerage for a firm in year t . In columns (2) and (4), the dependent variable, *Analyst-Forecast Error*, is the natural logarithm of 1 plus the absolute difference between the EPS forecast and actual EPS, divided by the share price at the beginning of the year and multiplied by 100. Standard errors clustered at the firm level are reported in parentheses. The superscripts ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

	# of Analyst Site Visits		Analyst-Forecast Error	
	(1)	(2)	(3)	(4)
HSR*Soft information	0.033** (0.016)		-0.025* (0.013)	
HSR* High return volatility		0.024 (0.026)		-0.034* (0.018)
HSR	0.036** (0.017)	0.037* (0.006)	-0.005 (0.012)	0.003 (0.006)
Control variables	Yes	Yes	Yes	Yes
Broker-firm pair FE	Yes	Yes	Yes	Yes
Firm city $\times$ Year FE	Yes	Yes	Yes	Yes
Brokerage city $\times$ Year FE	Yes	Yes	Yes	Yes
Observations	26,146	28,361	135,767	145,112
R-squared	0.569	0.553	0.633	0.611

5.8 Table 6: Two-stage least squares elasticity of outcomes to site visits

Regression	Coefficient	Interpretation
First stage: HSR $\rightarrow$ site visits	0.066***	HSR raises visits
Second stage: visits $\rightarrow$ forecast error	-0.359**	1% more visits reduces errors by 0.36%
Second stage: visits $\rightarrow$ FRINFO	0.456*	1% more visits raises forecast-revision reaction by 0.46%
Second stage: visits $\rightarrow$ recommendation reaction	0.409**	1% more visits raises recommendation reaction by 0.41%

Read. Using HSR as an instrument for site visits, more visits reduce forecast errors and increase market reactions to analyst outputs.

Economic message. The channel is direct information acquisition. HSR changes site visits, and site visits improve analyst output and price discovery.

5.9 Table 7: Soft information heterogeneity

Heterogeneity test	Site visits	Forecast errors
HSR $\times$ Soft information	0.033**	-0.025*
HSR $\times$ High return volatility	0.024	-0.034*

Read. The HSR effect is stronger when soft information is important and when firms are more opaque, as proxied by high return volatility.

Economic message. The value of in-person visits lies partly in collecting soft information that is difficult to summarize, transmit, and process remotely.

Table 8

Survey of remote and mega-city analysts.

Mega-city analysts are those employed by a broker located in Beijing, Shanghai, Shenzhen, or Guangzhou. Others are classified as remote respondents. Geographical distribution is in Panel A. Panel B summarizes the responses to each preliminary question. Panels C and D summarize the responses to general and specific questions by remote respondents and mega-city respondents, respectively. For all questions in Panels C and D, we use a 5-point scale (1 = Strongly Disagree, 5 = Strongly Agree). We reverse the order for the 51 responses that use the alternative survey questions. The % Agree column represents the percent of respondents that agreed or strongly agreed with the statement. A t-test is used to determine if the mean response is statistically different from the neutral midpoint of 3. *, **, and *** denote statistical significance at the 10%, 5%, and 1% level, respectively.

Panel A: Geographical distribution of remote and mega-city analysts

City	Respondents	
	Number	Percent
Mega-City Respondents (N = 236)		
Beijing	118	50.00%
Shanghai	78	33.05%
Shenzhen, Guangdong	30	12.71%
Guangzhou, Guangdong	10	4.24%
Remote Respondents (N = 98)		
Wuhan, Hubei	53	54.08%
Chengdu, Sichuan	20	20.41%
Changsha, Hunan	13	13.27%
Qingdao, Shandong	3	3.06%
Jiaxing, Zhejiang	3	3.06%
Jiangyin, Jiangsu	3	3.06%
Xiamen, Fujian	1	1.02%
Harbin, Heilongjiang	1	1.02%
Jiangmen, Guangdong	1	1.02%

Panel B: The frequency of remote and mega-city analyst responses to analyst profile questions (N = 334)

Number of firms following < = 8 (Yes/No)		remote	Yes	No	mega-city	Yes	No
			91 (92.80%)	7 (7.14%)		126 (53.39%)	110 (46.61%)
Direct HSR connection between brokerage and listed firm (Yes/No)			Yes	No		Yes	No
		remote	78 (79.59%)	20 (20.41%)	mega-city	170 (72.03%)	66 (27.97%)
Visit local firms more than nonlocal firms (Yes/No)			Yes	No		Yes	No
		remote	83 (84.69%)	15 (15.31%)	mega-city	160 (67.8%)	76 (32.2%)
Number of visits per year paid to portfolio firms			#visits = 0	1-3	3-5	#visits = 0	#visits = 3-5
		remote	6 (6.12%)	55 (56.12%)	29 (29.59%)	12 (5.08%)	121 (51.27%)
Industry distribution of portfolio firms (Manufacturing, High-tech, and Other industries)		mega-city	160 (67.8%)	Manufacturing	High-tech	Other industries	43 (18.22%)
		remote	48 (48.98%)	34 (34.69%)	16 (16.33%)	101 (42.8%)	57 (24.15%)
		mega-city	101 (42.8%)	57 (24.15%)	78 (33.05%)		

Panel C: Responses from remote respondents (N = 98)

General questions:			
Direct HSR connection results in analysts...	% Agree	Mean	St. Dev.
Conducting more site visits in person	95.92%	4.59***	0.61
Obtaining more firm-specific information	92.86%	4.47***	0.72
Better understanding strategies, operation, and performance	88.78%	4.37***	0.8
Better understanding key challenges/issues facing companies	86.73%	4.34***	0.9
Better understanding corporate culture and employees' morale	87.76%	4.35***	0.9
Specific questions:			
The introduction of a direct HSR route will...			
Increase your frequency of visit	91.84%	4.41***	0.87
Increase your flexibility to visit when most useful	95.92%	4.52***	0.76
Help you better understand current strategies, operation, performance, corporate culture, and employees' morale	86.73%	4.32***	0.84

(continued on next page)

Table 8

(continued)

Panel C: Responses from remote respondents (N=98)

General questions:			
Direct HSR connection results in analysts...	% Agree	Mean	St. Dev.
Enable you to talk to non-management employees to gain more insights into the company's strategies, operations, culture, and employment morale	88.78%	4.37***	0.83
Help you make more accurate earnings forecasts	81.63%	4.18***	0.99
Help you forecast the company's long-term growth	77.55%	4.03***	1.16

Panel D: Responses from mega-city respondents (N = 236)

General questions:			
Direct HSR connection results in analysts...	% Agree	Mean	St. Dev.
Conducting more site visits in person	86.86%	4.13***	0.79
Obtaining more firm-specific information	77.54%	3.93***	0.84
Better understanding strategies, operation, and performance	67.80%	3.77***	0.92
Better understanding key challenges/issues facing companies	65.25%	3.71***	0.92
Better understanding corporate culture and employees' morale	74.15%	3.86***	0.84
Specific questions:			
The introduction of a direct HSR route will...			
Increase frequency of visit	83.47%	4.10***	0.8
Increase flexibility to visit when most useful	89.83%	4.19***	0.76
Better understand current strategies, operation, performance, corporate culture, and employees' morale	66.10%	3.77***	0.9
Talk to non-management employees to gain more insights into the company's strategies, operations, culture, and employment morale	73.31%	3.88***	0.87
Make more accurate earnings forecasts	48.73%	3.44***	1.04
Help forecast the company's long-term growth	46.19%	3.37***	1.09

5.10 Table 8 and Figure 3: Survey evidence

Remote analysts' response	Percent agreeing	Mean response
HSR leads to more site visits	96%	4.59
HSR increases own visit frequency	91.84%	4.41
HSR increases flexibility to visit when useful	95.92%	4.52
HSR helps make more accurate earnings forecasts	81.63%	4.18
HSR helps forecast long-term growth	77.55%	4.03

Read. The survey strongly supports the empirical mechanism. Remote analysts report that HSR increases visits, makes visits more flexible, helps them collect firm-specific information, and improves forecast accuracy.

Economic message. The survey evidence is useful because it comes directly from the agents whose behavior the paper studies. It also clarifies the type of information acquired: soft information about strategy, operations, challenges, culture, and morale.

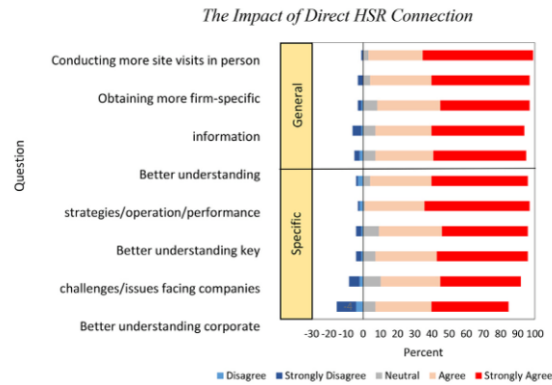
5.11 Table 9: HSR introduction and coverage initiation

Outcome: coverage initiation	All firms	SZSE firms after 2009
HSR	0.006***	0.006***

Read. HSR increases the probability that a broker initiates coverage of a newly connected firm by

Panel A: Survey responses among remote respondents

This figure shows the distribution of combined responses to the original and alternative survey questions provided in Internet Appendix IA.5 and IA.6, respectively, among remote analysts. Remote analysts are those employed by a broker located in cities other than Beijing, Shanghai, Shenzhen, or Guangzhou. The first five rows show the responses to the general questions, and the last six rows for the specific questions. On the horizontal axis, positive (negative) percentages refer to "strongly agree," "agree," and "neutral" ("disagree" and "strongly disagree") responses.



Panel B: Survey responses among mega-city respondents

This figure shows the distribution of responses to the original survey questions provided in Internet Appendix IA.5 among mega-city analysts. Mega-city analysts are those employed by a broker located in Beijing, Shanghai, Shenzhen, or Guangzhou. The first five rows show the responses to the general questions, and the last six rows show the responses to the specific questions. On the horizontal axis, positive (negative) percentages refer to "strongly agree," "agree," and "neutral" ("disagree" and "strongly disagree") responses.

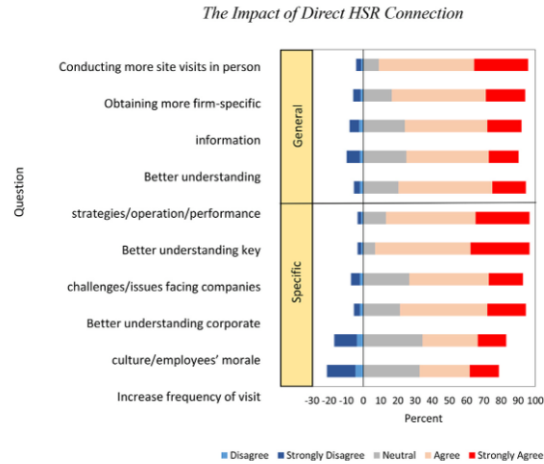


Fig. 3. Survey responses.

about 0.6 percentage points. This is about 14% of the unconditional mean initiation rate of 4.2 percentage points.

Economic message. Lower acquisition costs affect both the intensive margin and the extensive margin. Analysts not only visit already covered firms more; they are also more likely to begin covering new firms.

Table 9

HSR introduction and analyst coverage-initiation decision.

This table reports the results of testing the effect of HSR introduction on analyst coverage-initiation decision. We start with all firms that are included in the CSMAR database. For a broker-firm-year to be included in the coverage-initiation sample we require that the broker issues no earnings forecast in the prior three years from $t-3$ to $t-1$. The dependent variable, *Initiation*, is a dummy variable equal to 1 if a broker issues a forecast for firm i in year t , and 0 otherwise. *HSR* is an indicator variable that takes the value of 1 if at least one HSR connects the firm city and the brokerage city at the beginning of t , and 0 otherwise. Control variables include firm size, leverage, ROA, book-to-market, and return volatility. All variables are defined in [Appendix A.Table A1](#). We include broker-firm-pair fixed effects, firm city-by-year fixed effects, and broker city-by-year fixed effects. Standard errors clustered at the firm level are reported in parentheses. The superscripts ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

	<i>Initiation</i>	
	All Firms	SZSE firm after 2009
	(1)	(2)
HSR	0.006*** (0.001)	0.006*** (0.002)
Control variables	Yes	Yes
Broker-firm pair FE	Yes	Yes
Firm city $\times$ Year FE	Yes	Yes
Brokerage city $\times$ Year FE	Yes	Yes
Observations	549,607	281,238
R-squared	0.337	0.362

Table 10

Cross-sectional variation of the HSR effect on forecast errors in corporate governance.

This table examines whether the effect of HSR introduction between a firm-brokerage pair on analyst-forecast error varies with firm corporate governance. Two indicator variables serve as the proxy for corporate governance. *SOE* takes the value of 1 if the firm is a state-owned enterprise in the year immediately before the HSR introduction, and 0 otherwise. *Low board independence* takes the value of 1 if the proportion of the independent board of directors is below the sample median in the year immediately before the HSR introduction, and 0 otherwise. The dependent variable, *Analyst-Forecast Error*, is the natural logarithm of one plus the absolute difference between the EPS forecast and actual EPS, divided by the share price at the beginning of the year and multiplied by 100. *HSR* is an indicator variable that takes the value of 1 if at least one HSR connects the firm city and the brokerage city at the beginning of t , and 0 otherwise. Firm-broker-pair fixed effects and city-times-year fixed effects are included as controls. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels, respectively.

	<i>Analyst-Forecast Error</i>	
	(1)	(2)
HSR*SOE	-0.056** (0.026)	
HSR*Low board independence		-0.083** (0.038)
HSR	0.013 (0.017)	-0.006 (0.010)
Control variables	Yes	Yes
Pair FE	Yes	Yes
Firm city $\times$ Year FE	Yes	Yes
Brokerage city $\times$ Year FE	Yes	Yes
Observations	141,104	145,418
R-squared	0.614	0.611

5.12 Table 10: Corporate governance heterogeneity

Outcome: forecast error	Coefficient	Interpretation
HSR $\times$ state-owned enterprise	-0.056**	Stronger forecast-error reduction for SOEs
HSR $\times$ low board independence	-0.083**	Stronger effect for weak boards

Read. The forecast-quality improvement is stronger for firms with weaker governance, proxied by state ownership and low board independence.

Economic message. When managers have weaker incentives to disclose information voluntarily, analyst visits become more valuable. HSR helps analysts overcome opacity and extract value-relevant information.

6 Short summary

Chen, Ma, Martin, and Michaely study whether lower information-acquisition costs lead to more information production, better information quality, and more efficient prices. Their setting is the expansion of China's high-speed rail network, which lowers the cost of analysts traveling to visit portfolio firms.

The paper uses firm-brokerage-pair variation in HSR connectivity. A pair becomes treated when an HSR route connects the brokerage city and the firm city. The authors compare the same pair before and after connection, while controlling for pair fixed effects and city-year shocks. They also use specifications with firm-year and broker-year fixed effects.

The first main result is that HSR increases analyst site visits. The baseline estimate implies about a 4.9% increase in site visits, and the effect is stronger for firm-broker pairs that were difficult to visit before HSR.

The second main result is that HSR improves forecast quality. Forecast errors fall by about 2.1% in the baseline model. Again, the effect is stronger for difficult-to-visit pairs, suggesting the improvement operates through easier information acquisition.

The third main result is that HSR improves price formation. Market reactions to analyst forecast revisions and recommendations become stronger, while market reactions to earnings announcements fall when a larger share of analysts are HSR-connected to the firm. The interpretation is that analysts reveal more information earlier, so prices incorporate information faster.

The fourth main result is about mechanism. The effect is strongest when soft information matters. Survey evidence from financial analysts confirms that HSR increases site visits, improves travel flexibility, and helps analysts understand firm-specific soft information such as strategy, operations, corporate culture, and employee morale.

Overall, the paper is a clean empirical test of a foundational assumption in information economics: cheaper information acquisition leads agents to acquire more information, produce higher-quality output, and improve the efficiency of market outcomes.

7 Conclusion

7.1 Core conclusion

The paper finds that reducing information-acquisition costs increases analysts' information production, improves the quality of their forecasts, and helps stock prices incorporate information more efficiently.

7.2 Economic intuition

Analysts can produce valuable information, but doing so requires costly effort. When HSR lowers the cost of visiting firms, analysts make more visits. These visits help them collect soft information that is hard to obtain remotely. Better information improves forecast accuracy, makes analyst research more useful to investors, and reduces the amount of surprise left for earnings announcements.

7.3 Main empirical contribution

The central contribution is that the paper directly observes information collection through site visits. Many studies infer information acquisition from outcomes. This paper links the cost shock, the acquisition activity, the quality of analyst output, and the price-efficiency consequences in one empirical framework.

7.4 Why the paper matters

The results matter for financial economics because they validate a key assumption behind models of costly information acquisition. They also matter for policy because transportation infrastructure can create informational externalities in financial markets. Infrastructure does not only move people and goods; it can also move information into prices.

7.5 Takeaway

High-speed rail put analysts on the fast track to firms. By lowering the cost of face-to-face information collection, it increased analyst visits, improved forecast accuracy, strengthened market reactions to analyst research, and made earnings announcements less surprising.